BPSK SYSTEM ARCHITECTURE & VERIFICATION REPORT

EECE 5498: Data-Driven Decision Making & Risk Management — Assignment 2
Bayesian Decision Boundaries, SysML Traceability & Monte Carlo Error Analysis

1. EXECUTIVE CONTEXT & SYSTEM SPECIFICATIONS

This report documents the architectural design, SysML modeling, and empirical performance verification for a Binary Phase Shift Keying (BPSK) digital communication system under additive white Gaussian noise (AWGN). Prepared for **EECE 5498: Data-Driven Decision Making and Risk Management** at UMass Lowell, this work evaluates **Assignment 2: On Bayesian Decision Thresholds and Error Probabilities**.

Req ID	Requirement Name	Category	Engineering Specification
REQ_BPSK_01	**Bit Error Rate Constraint**	Performance	Empirical BER shall be less than or equal to 0.01 (BER <= 0.01) under baseline operating noise conditions.
REQ_BPSK_02	**Maximum Transmit Power**	Design Constraint	Transmitter peak signal power shall not exceed 4.0 W (P = A^2 <= 4.0 W), requiring peak amplitude A <= 2.0 V.
REQ_BPSK_03	**Antipodal Signaling Format**	Functional	System shall implement BPSK transmitting equiprobable binary symbols (+A, -A) at 1 bit per symbol.
REQ_BPSK_04	**Optimal Decision Threshold**	Physical / Logic	Receiver shall classify noisy signals using an optimal zero-voltage decision boundary (xr = 0 V).

2. SYSML ARCHITECTURE & SYSTEM DECOMPOSITION

The system architecture establishes full traceability between requirements, structural blocks, port connections, and operational logic using Model-Based Systems Engineering (MBSE) in SysML:

• **Block Definition Diagram (BDD)**: Structural breakdown into three core components — :tx BPSK Transmitter, :channel Channel Noise, and :rx BPSK Receiver under the top-level BPSK System context.

• **Internal Block Diagram (IBD)**: Maps directional port flow: tx.tx_out → channel.channel_in → channel.channel_out → rx.rx_in.

• **3-Swimlane Activity Diagram**: Allocates execution logic across transmitter (Bernoulli symbol generation), channel (Gaussian noise addition: mu_N = 0, sigma_N^2 = 0.5), and receiver (decision thresholding & BER computation).

> **SysML Requirement Allocation:** REQ_BPSK_02 and REQ_BPSK_03 are allocated to the Transmitter block (A = 2.0 V, Power = 4.0 W). REQ_BPSK_01 and REQ_BPSK_04 are allocated to the Receiver block (xr = 0 V, BER <= 0.01). All dependencies are formally linked via «satisfy» relationships.

3. BAYESIAN DECISION BOUNDARY DERIVATION

The receiver decision boundary **xr = 0.0 V** is derived from Bayesian decision theory and average risk minimization. For antipodal BPSK signaling, received signals represent two Gaussian conditional likelihoods: p(r | w1) ~ N(-A, sigma_N^2) for symbol -A, and p(r | w2) ~ N(+A, sigma_N^2) for symbol +A.

> **Bayes Minimum Risk Derivation:**
> • **Symmetric Loss Matrix (L):** Zero penalty for correct decisions (lambda_11 = lambda_22 = 0) and equal misclassification loss (lambda_12 = lambda_21 = 1.0).
> • **Equiprobable Priors:** Binary symbols occur with equal likelihood (P(w1) = P(w2) = 0.5).
> • **Threshold Formula:** xr = (mu_1 + mu_2)/2 + (sigma_N^2 / (mu_2 - mu_1)) · ln([lambda_12 · P(w1)] / [lambda_21 · P(w2)])
> • **Result:** Substituting mu_1 = -A, mu_2 = +A, and ln(1.0) = 0 yields **xr = 0.0 V**.
> This proves that zero-voltage thresholding is the mathematically exact Bayes-optimal decision boundary.

4. MONTE CARLO SIMULATION & ANALYTICAL VERIFICATION

To verify system compliance, Monte Carlo simulations were executed with amplitude **A = 2.0 V**, noise variance **sigma_N^2 = 0.5 V^2**, and trial count **N_bit = 1000**. Theoretical performance is governed by the complementary error function (erfc):

Parameter Metric	Theoretical Analytical Value	Simulated Empirical Value	Requirement Limit	Status
Operating SNR	6.02 dB (SNR_linear = 4.0)	6.02 dB	N/A	**Calculated**
Transmit Power (P)	4.00 W (A = 2.0 V)	4.00 W	<= 4.00 W (REQ_BPSK_02)	**PASS**

Decision Boundary (xr)	0.00 V (Bayes Optimal)	0.00 V	xr = 0.00 V (REQ_BPSK_04)	**PASS**
Bit Error Rate (BER)	0.00234 (0.234%)	0.00200 (2 / 1000 bits)	<= 0.01000 (REQ_BPSK_01)	**PASS**

As shown in the simulated BER vs. SNR waterfall curve (ber_graph.png), the empirical data point (BER = 0.0020 at 6.02 dB) lies directly on the theoretical curve, comfortably below the horizontal BER = 0.01 constraint line.

5. FINAL REQUIREMENTS VERIFICATION MATRIX

All four formal engineering requirements specified for Assignment 2 have been fully satisfied:

Req ID	Description	Target Requirement	Observed Result	Verification Outcome
REQ_BPSK_01	**Bit Error Rate Constraint**	BER <= 0.0100	BER = 0.0020	**SATISFIED**
REQ_BPSK_02	**Maximum Transmit Power**	P <= 4.0 W (A <= 2.0 V)	P = 4.0 W (A = 2.0 V)	**SATISFIED**
REQ_BPSK_03	**Antipodal Signaling Format**	1 bit/symbol (+A, -A)	+A / -A mapped (pBer = 0.5)	**SATISFIED**
REQ_BPSK_04	**Optimal Decision Threshold**	xr = 0.0 V	xr = 0.0 V (Bayes Optimal)	**SATISFIED**

Summary Conclusion: The BPSK system architecture and simulation model meet 100% of the specified engineering requirements for EECE 5498 Assignment 2. The SysML structural models (BDD, IBD) and 3-swimlane activity diagram establish complete traceability, while the empirical Monte Carlo BER (0.0020) closely matches the analytical Q-function prediction (0.00234), confirming robust design performance.